AG News Classification using Machine Learning and Natural Language Processing

By

Name: - Nimit Singh
Enrolment No: - 202427600018

A Report Submitted to
Gujarat University
In Partial Fulfilment of the Requirements
for the Degree of **M.Sc. IT in**
Business Intelligence and Analytics
November, 2025

Centre for Professional Courses,

Gujarat University, Ahmedabad

CERTIFICATE

This is to certify that research work embodied in this report entitled "**AG News Classification using Machine Learning and Natural Language Processing**" was carried out by **Nimit Singh (Enrolment No: 202427600018**) at Centre for Professional Course for partial fulfilment of M.Sc. IT in Business Intelligence and Analytics degree to be awarded by Gujarat University. This research work has been carried out under my supervision and is to the satisfaction of department.

Date: 30/06/2026

Place: Ahmedabad

Nilesh Gol

Assistant Professor

(Guide)

CPC, Gujarat University

Amandeep Kaur

Program In-Charge

CPC, Gujarat University

Dr. Paavan Pandit

Director

CPC, Gujarat University

Seal of Institute

DECLARATION OF ORIGINALITY

I hereby certify that I am the sole author of this Project report and that neither any part of this Project report nor the whole of the Project report has been submitted for a degree to any other University or Institution.

I certify that, to the best of my knowledge, my Project report does not infringe up on any one's copyright nor violate any proprietary rights and that any ideas, techniques, quotations, or any other material from the work of other people included in my Project report, published or otherwise, are fully acknowledged in accordance with the standard referencing practices.

I declare that this is a true copy of my Project report, including any final revisions, as approved by my Project report review committee.

Date: 29/03/26
Place: Ahmedabad

Name: - Nimit Singh
Enrolment no: - 202427600018

PROJECT REPORT APPROVAL

This is to certify that research work embodied in this Project report entitled "**AG News Classification using Machine Learning and Natural Language Processing**" was carried out by **Nimit Singh (Enrolment No: 202427600018)** at Centre for Professional Course for partial fulfilment of M.Sc. IT degree in Business Intelligence and Analytics to be awarded by Gujarat University.

Date: 29/03/26
Place: Ahmedabad

Examiner(s):

(_____) (_____)

(_____)

ACKNOWLEDGEMENT

We are sincerely thankful to our guide, for their constant support, stimulating suggestions, and encouragement, which greatly assisted us in successfully completing our project work. Their close supervision over the past few months and helpful insights have been invaluable. Despite their busy schedule, their valuable advice and unwavering support have been an inspiration and a driving force for us. Their experience and knowledge have continuously helped shape our initial ideas into a comprehensive form.

I, hereby, take an opportunity to convey my gratitude for the generous assistance and cooperation, that I received from the **Amandeep Kaur** and to all those who helped me directly and indirectly.

We are deeply indebted & thankful to our Department Faculties who helped and rendered their valuable time, knowledge and information and whose suggestion and guidance has enlightened on the subject.

We also thank **"Dr. Paavan Pandit,"** Director, CPC, GU for extending all the help and cooperation during our training period.

Finally, I am also indebted to my friends without whose help I would have had a hard time managing everything on my own.

Nimit Singh

(Enrolment No: 202427600018)

Table of Contents

Chapter1 – Introduction

1.1 Overview
1.2 Problem Definition
1.3 Objectives
1.4 Motivation
1.5 Scope of the Project
1.6 Applications

Chapter 2 – Literature Review

2.1 Introduction
2.2 Related Work
2.3 Machine Learning in Prediction
2.4 Linear Regression Model
2.5 Summary

Chapter 3 – System Analysis and Dataset

3.1 Dataset Description
3.2 Variables
3.3 Exploratory Data Analysis (EDA)
3.4 Data Preprocessing

Chapter 4 System Design

4.1 Introduction
4.2 System Architecture
4.3 Model Design
4.4 Performance Metrics
4.5 User Interface (Streamlit App)
4.6 Summary

Chapter 5 – Implementation

5.1 Libraries
5.2 Model Implementation Steps
5.3 Code Integration
5.4 Deployment

Chapter 6 – Results and Discussion

6.1 Performance Metrics
6.2 Visualization

6.3 Streamlit App Output

Chapter 7 – Conclusion and Future Scope

7.1 Summary
7.2 Applications
7.3 Future Work

Chapter 8 – Bibliography & References

8.1 Books & Research Papers
8.2 Websites

ABSTRACT

The project titled “AG News Classification using Machine Learning and Natural Language Processing” focuses on automatically classifying news articles into predefined categories such as World, Sports, Business, and Science/Technology. With the rapid growth of digital news content, manual classification has become inefficient and time-consuming. This project aims to address this problem by developing an intelligent text classification system using machine learning techniques.

The dataset used for this project is the AG News dataset, which consists of labeled news articles. The textual data is preprocessed using Natural Language Processing (NLP) techniques such as lowercasing, removal of punctuation, and tokenization. The processed text is then converted into numerical form using TF-IDF vectorization, enabling the machine learning model to understand and learn patterns from the data.

A Logistic Regression algorithm is used as the classification model due to its efficiency and strong performance on text-based data. The model is trained on the training dataset and evaluated on the test dataset, achieving reliable accuracy in predicting the correct category of news articles.

To enhance usability, a Streamlit-based web application is developed, allowing users to input news text and instantly receive its predicted category. This project demonstrates the practical application of machine learning and NLP in solving real-world problems such as content organization, news filtering, and recommendation systems.

Chapter 1 - Introduction

1.1 Overview

In today's digital era, a massive amount of news content is generated every day across various platforms such as websites, mobile applications, and social media. Organizing and categorizing this information manually is a time-consuming and inefficient process. To address this challenge, automated news classification systems are developed using Machine Learning (ML) and Natural Language Processing (NLP) techniques.

The project "AG News Classification using Machine Learning and Natural Language Processing" aims to automatically classify news articles into predefined categories such as World, Sports, Business, and Science/Technology. The system analyzes the textual content of news articles and predicts the most appropriate category based on learned patterns from the dataset.

1.2 Problem Definition:

With the rapid growth of online news platforms, managing and organizing large volumes of textual data has become increasingly difficult. Manual classification is not only slow but also prone to human error.
The main problem addressed in this project is:

- To automatically classify news articles into appropriate categories based on their textual content using machine learning techniques.

1.3 Objectives:

The key objectives of this project are:

- To develop an automated system for classifying news articles
- To apply Natural Language Processing techniques for text preprocessing
- To convert textual data into numerical form using vectorization techniques
- To build and train a machine learning model (Logistic Regression)
- To evaluate the model's performance on test data
- To develop a user-friendly Streamlit web application for real-time predictions

1.4 Motivation:

The increasing volume of digital news content has created a need for efficient and intelligent systems that can organize information automatically. Accurate classification of news articles improves user experience, enables faster information retrieval, and supports recommendation systems.

This project is motivated by the need to:

- Reduce manual effort in news categorization
- Improve accuracy and efficiency
- Apply machine learning in real-world applications

1.5 Scope of the Project:

The scope of this project includes:

- Classification of English news articles into four categories
- Implementation of NLP preprocessing techniques
- Use of machine learning for prediction
- Development of a web-based interface using Streamlit
- The system can be extended in the future to support more categories, languages, and advanced models.

1.6 Applications:

The developed system can be used in various real-world applications, including:

- News Websites: Automatic categorization of articles
- Search Engines: Improved indexing and search results
- Content Recommendation Systems: Personalized news suggestions
- Social Media Platforms: Organizing and filtering content
- Data Analytics: Trend analysis based on categorized news

Chapter 2: Literature Review

2.1 Introduction:

This chapter reviews existing research and methodologies related to text classification and the application of Machine Learning (ML) and Natural Language Processing (NLP) in this domain. It highlights how different techniques have been used to automatically classify textual data and justifies the choice of methods used in this project.

2.2 Related Work:

Text classification has been widely studied in the field of machine learning. Various researchers have explored different algorithms and approaches to improve classification accuracy and efficiency.

Traditional methods such as Naive Bayes, Support Vector Machines (SVM), and Decision Trees have been commonly used for text classification tasks. These models have shown good performance, especially when combined with feature extraction techniques like Bag-of-Words and TF-IDF.

Recent advancements have also introduced deep learning approaches such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Transformer-based models (like BERT), which provide higher accuracy but require more computational resources.

2.3 Machine Learning in Text Classification:

Machine Learning plays a crucial role in automatically analyzing and classifying textual data. It enables systems to learn patterns from large datasets and make predictions on new, unseen data.

In text classification, the process typically involves:

- Preprocessing the text data
- Converting text into numerical features
- Training a classification model
- Evaluating performance

Machine learning models are effective because they can generalize patterns and adapt to different types of textual content.

2.4 Natural Language Processing Techniques:

Natural Language Processing (NLP) is used to prepare and transform raw text into a format suitable for machine learning models. Some commonly used NLP techniques include:

- Tokenization: Splitting text into words or tokens
- Lowercasing: Converting all text to lowercase
- Stopword Removal: Removing common words like “the”, “is”, “and”
- Stemming/Lemmatization: Reducing words to their root form
- Vectorization: Converting text into numerical format using methods like TF-IDF

In this project, TF-IDF (Term Frequency–Inverse Document Frequency) is used to represent textual data numerically.

2.5 Logistic Regression for Classification:

Logistic Regression is a widely used machine learning algorithm for classification problems. It is particularly effective for text classification tasks due to its simplicity, speed, and ability to handle high-dimensional data.

The model estimates the probability of a data point belonging to a particular class and assigns it accordingly. Despite being a simple algorithm, Logistic Regression often performs competitively compared to more complex models when applied to text data.

2.6 Summary

The literature review highlights that machine learning and NLP techniques are highly effective for text classification tasks. While advanced deep learning models offer higher accuracy, traditional models like Logistic Regression provide a good balance between performance and efficiency.

Based on this analysis, Logistic Regression combined with TF-IDF vectorization is chosen for this project as it delivers reliable results with lower computational complexity.

Chapter 3 – System Analysis and Dataset

3.1 Dataset Description:

The dataset used in this project is the AG News Dataset, which is widely used for text classification tasks. It consists of news articles categorized into four classes: World, Sports, Business, and Science/Technology.

The dataset is divided into two parts:

- train.csv → Used for training the model
- test.csv → Used for evaluating the model

Each record in the dataset contains:

- Class Index (Label): Represents the category of the news
- Title: Headline of the news article
- Description: Detailed content of the news

The dataset is balanced, meaning each category has approximately an equal number of samples, which helps in building an unbiased model.

3.2 Variables:

Type	Variable Name	Description
Input	Title	News headline
Input	Description	Full news content
Output	Label	Category (1–4)

Category Mapping:

Label	Category
1	World
2	Sports
3	Business
4	Sci/Tech

3.3 Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) is performed to understand the dataset and identify patterns.

Key Analysis Performed:

- Checked dataset structure and data types
- Verified class distribution
- Observed text length and patterns

Insights:

- The dataset is well-balanced across all categories
- News titles and descriptions vary in length
- Each category contains distinct keywords

Example Observations:

- Sports news includes words like match, team, win
- Business news includes market, company, stock
- Sci/Tech includes technology, software, internet

These patterns help the model learn differences between categories.

3.4 Data Preprocessing:

Data preprocessing is an important step to prepare text for machine learning.

Steps Performed:

- Lowercasing: Converted all text to lowercase
- Removing punctuation: Cleaned unwanted symbols
- Tokenization: Split text into words
- Combining Text: Title and description merged (if applied)

Feature Extraction (Vectorization)

The cleaned text is converted into numerical form using:

TF-IDF (Term Frequency – Inverse Document Frequency)

This technique:

- Assigns importance to words
- Reduces impact of common words
- Helps model understand meaningful patterns

Train-Test Split

- Training Data → Used to train the model
- Testing Data → Used to evaluate performance

3.5 Summary

This chapter described the dataset structure, variables, and preprocessing steps used in the project. Proper data preparation and feature extraction are essential for achieving good model performance. The AG News dataset provides a strong foundation for building an accurate text classification model.

Chapter 4 – System Design

4.1 Introduction

System Design explains how the overall system is structured and how different components interact to perform news classification. It includes the data flow, model design, performance metrics, and the user interface used for prediction.

The goal of this system is to provide an efficient, accurate, and user-friendly solution for classifying news articles automatically.

4.2 System Architecture

The architecture of the system follows a step-by-step pipeline:

User Input (News Text)
↓
Text Preprocessing
↓
Feature Extraction (TF-IDF)
↓
Machine Learning Model (Logistic Regression)
↓
Predicted Category Output

Explanation of Components:

1. **User Input:**
   The user enters a news article (title or description) through the Streamlit interface.
2. **Text Preprocessing:**
   The input text is cleaned by converting to lowercase, removing punctuation, and tokenizing.
3. **Feature Extraction:**
   The cleaned text is converted into numerical form using TF-IDF vectorization.
4. **Machine Learning Model:**
   The trained Logistic Regression model processes the input and predicts the category.
5. **Output:**
   The predicted category (World, Sports, Business, Sci/Tech) is displayed to the user.

4.3 Model Design:

The system uses Logistic Regression as the classification model.

Working Principle:

- Logistic Regression calculates the probability of input text belonging to each category.
- The category with the highest probability is selected as the final output.

Why Logistic Regression?

- Efficient for large text datasets
- Works well with TF-IDF features
- Fast training and prediction
- Easy to interpret

4.4 Performance Metrics

To evaluate the model, the following metrics are used:

Metric	Description
Accuracy	Percentage of correct predictions
Precision	Correct positive predictions
Recall	Ability to find all relevant cases
F1 Score	Balance between precision and recall

These metrics help measure how well the model performs on unseen data.

4.5 User Interface (Streamlit App)

A web application is developed using Streamlit to make the system interactive and easy to use.

Features of the App:

- Input text box for entering news articles
- “Classify” button to trigger prediction

- Displays predicted category with labels:
  - World
  - Sports
  - Business
  - Sci/Tech

Advantages:

- Simple and clean interface
- Real-time prediction
- No coding required for users

4.6 Summary

This chapter described the design of the system, including the data flow, model structure, evaluation metrics, and user interface. The system is designed to be efficient, scalable, and user-friendly, making it suitable for real-world applications.

Chapter 5 – Implementation

5.1 Introduction:

This chapter explains the practical implementation of the news classification system. It covers the tools and libraries used, step-by-step model development, integration of components, and deployment of the application using Streamlit.

5.2 Libraries Used:

The following Python libraries are used in this project:

Library	Purpose
Python	Programming language
Pandas	Data manipulation and analysis
NumPy	Numerical operations
Scikit-learn	Machine learning model and TF-IDF
Streamlit	Web application development
Joblib	Model saving and loading

5.3 Model Implementation Steps:

Step 1: Load Dataset

- The dataset (train.csv and test.csv) is loaded using Pandas
- Data is checked for structure and consistency

Step 2: Data Preprocessing

- Converted text to lowercase
- Removed punctuation and unwanted characters
- Combined title and description (if applied)
-

Step 3: Feature Extraction

- Used TF-IDF Vectorizer to convert text into numerical form

- Helps the model understand important words

Step 4: Model Training

- Logistic Regression model is trained using training data
- The model learns patterns from text and categories

Step 5: Model Evaluation

- Model tested on test dataset
- Performance measured using accuracy and other metrics

Step 6: Model Saving

- The trained model is saved using Joblib
- File name: News_Classificatiom_Model_LR.joblib

5.4 Code Integration:

The trained model is integrated into a Streamlit web application.

Key Implementation (from your code):

- Model loaded using joblib.load()
- User input taken using st.text_area()
- Prediction triggered using button click
- Output displayed using st.success()

Category Mapping:

- 1 → World
- 2 → Sports
- 3 → Business
- 4 → Sci/Tech

5.5 Deployment:

The application is deployed using Streamlit.

Application Details:

- App Name: News Classification App
- Framework: Streamlit
- Input: News article text
- Output: Predicted category

Features:

- User-friendly interface
- Real-time predictions
- Lightweight and fast

5.6 Summary:

This chapter described the complete implementation process, from data loading and preprocessing to model training, saving, and deployment. The integration of machine learning with a Streamlit interface makes the system practical and easy to use.

Chapter 6 – Results and Discussion

6.1 Introduction:

This chapter presents the results obtained from the machine learning model and discusses its performance. It also includes observations from the dataset and the effectiveness of the Streamlit application.

6.2 Model Performance Metrics:

The performance of the model is evaluated using standard classification metrics:

Metric	Description
Accuracy	Percentage of correctly classified news articles
Precision	Correct positive predictions for each class
Recall	Ability to correctly identify all relevant instances
F1 Score	Harmonic mean of precision and recall

Performance Analysis:

- The model performs well on unseen data
- It correctly identifies most of the news categories
- Logistic Regression provides stable and reliable results

(If you give me your exact accuracy, I will insert it here like: Accuracy = 88.7% — very important for marks)

6.3 Visualization and Analysis:

To better understand the model and dataset, different analyses are performed:

1. Class Distribution
    - The dataset is balanced across all categories
    - This helps in reducing bias in predictions
2. Text Pattern Analysis
    - Each category contains unique keywords
    - Example:
    - Sports → match, team, win
    - Business → market, company
    - Sci/Tech → software, internet

3. Model Behavior

- Performs well for clear and distinct text
- Slight confusion may occur in overlapping topics

6.4 Streamlit Application Output:

The developed Streamlit web application allows users to test the model in real time.
Working:

1. User enters news text
2. Clicks on “Classify” button
3. Model predicts category

Sample Output:

- Input: “The stock market showed a significant rise today…”
- Output: Business

Advantages of the App:

- Easy to use
- Instant prediction
- Interactive interface

6.5 Discussion:

- The model successfully classifies news articles with good accuracy
- Logistic Regression proves effective for text classification
- TF-IDF helps in identifying important words
- The system performs efficiently in real-time applications

However:

- Some categories may overlap (e.g., Business & Tech)
- Performance depends on input quality

6.6 Summary:

This chapter demonstrated that the developed model performs effectively in classifying news articles. The results show that machine learning combined with NLP techniques can solve real-world text classification problems efficiently.

Chapter 7 – Conclusion and Future Scope

7.1 Summary

The project "AG News Classification using Machine Learning and Natural Language Processing" successfully demonstrates how machine learning can be used to automatically classify news articles into predefined categories such as World, Sports, Business, and Science/Technology.

The system uses Natural Language Processing techniques for text preprocessing and TF-IDF vectorization for feature extraction. A Logistic Regression model is trained on the dataset and performs effectively in predicting the correct category of news articles.

Additionally, a Streamlit-based web application is developed to provide a user-friendly interface where users can input news text and get instant predictions. The system is efficient, accurate, and easy to use, making it suitable for real-world applications.

7.2 Applications

This project has several practical applications in different domains:

- News Websites: Automatic categorization of articles
- Search Engines: Improved indexing and search results
- Content Recommendation Systems: Personalized news suggestions
- Social Media Platforms: Filtering and organizing content
- Data Analytics: Analyzing trends in categorized news

7.3 Future Scope

The project can be further improved and extended in several ways:

1. Advanced Machine Learning Models

- Use models like:
    - Random Forest
    - XGBoost
    - Support Vector Machines

2. Deep Learning Techniques

- Implement advanced NLP models such as:
    - LSTM (Long Short-Term Memory)
    - BERT (Bidirectional Encoder Representations from Transformers)

3. Increase Dataset Size

- Use larger and more diverse datasets
- Improve model generalization and accuracy

4. Multilingual Support

- Extend the system to support multiple languages
- Make it globally applicable

5. Real-Time News Integration

- Connect with live news APIs
- Classify real-time news articles

6. Improved User Interface

- Enhance UI design
- Add features like history, analytics, and category filtering

7**.4 Final Conclusion:**

This project highlights the power of machine learning and NLP in solving real-world problems like text classification. The developed system provides a strong foundation for building more advanced and scalable applications in the future.

Chapter 8: References

8.1 Books & Research Papers

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurélien Géron, O'Reilly Media, 2022.
2. Python Data Science Handbook – Jake VanderPlas, O'Reilly Media, 2021.
3. "Text Classification using Machine Learning Techniques," International Journal of Computer Applications, 2020.
4. "A Survey on Text Classification Algorithms," IEEE Access Journal, 2021.
5. "Natural Language Processing with Machine Learning," Elsevier Publications, 2022.

8.2 Websites

1. Scikit-learn Official Documentation
   https://scikit-learn.org/
2. Streamlit Official Website
   https://streamlit.io/
3. Pandas Documentation
   https://pandas.pydata.org/
4. NumPy Documentation
   https://numpy.org/
5. Kaggle Dataset Source
   https://www.kaggle.com/